FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2021

PAPER –1

TEST DATE: 18-01-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. $I = \frac{mL^2}{3} + \frac{\frac{m}{2}\left(\frac{1}{2}\right)^2}{12} + \frac{m}{2}.\left(L^2 + \frac{L^2}{16}\right)$

$= \frac{mL^2}{3} + \frac{mL^2}{96} + \frac{mL^2}{2} + \frac{mL^2}{32} = \frac{7mL^2}{8}$

$\overrightarrow{r_{ar}} = \frac{m\left(\frac{L}{2}\hat{i}\right) + \frac{m}{2}\left(L\hat{i} + \frac{L}{4}\hat{j}\right)}{3m/2} = \frac{2}{3}\left(L\hat{i} + \frac{L}{8}\hat{j}\right)$

$\tau_{max} = \frac{3mg}{2} \times \frac{2}{3} \times L.\frac{\sqrt{65}}{8} = \frac{\sqrt{65}}{8}mgL$

$\alpha = \frac{\tau_{max}}{I}$

2. A

Sol. Basic concept.

3. C

Sol. AB and CD are isochoric process & BC and DA are isothermal process.

$W_{net} = W_{AB} + W_{BC} + W_{CD} + W_{DA}$

$$= 0 + W_{BC} + 0 + W_{DA}$$

$$= nRT_2 \ell n\left(\frac{V_C}{V_B}\right) + nRT_1 \ell n\left(\frac{V_A}{V_D}\right)$$

$$= -\frac{nM}{P_0}\ell\left(\frac{3P_0}{P_0}\right).(2P_0 - P_0)$$

$$= -\frac{nMP_0}{\rho_0}\ell n(3)$$

4. B

Sol. $-\frac{Q_0 + q}{C} - \frac{Ldi}{dt} + \frac{8Q_0 - q}{2C} = 0$

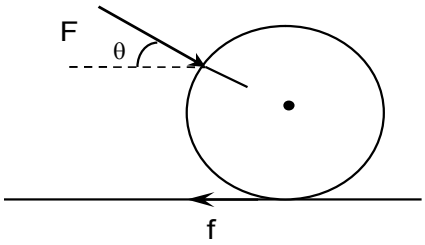
$\frac{di}{dt} = 0$ when current is maximum.

5. B

Sol. Basic concept. $I = \frac{p_0^2}{2pv}$

6. A

Sol. $F = \frac{I.\pi R^2}{C}$



$F\cos\theta - f = ma$

$fR = \frac{2}{5}MR^2(\alpha)$

$a = \alpha R$

7. D

Sol. Basic concept.

8. AC

Sol. Basic concept.

9. ABC

Sol. Dipole moment $= Q \times 2 \times \frac{5a}{2} \times \frac{1}{3} = \frac{5Qa}{3}$

$\vec{\tau} = \vec{p} \times \vec{E} \Rightarrow \tau = pE\sin\theta$

$U = -\vec{p}.\vec{E}$

10. ABCD

Sol. $E = -E_0 z^2 / n^2; v = v_0.z / n$ and $r = r_0.\frac{n^2}{z}$

11. ABCD

Sol. $\frac{Ldi}{dt} = 5t \quad \text{for } 0 \le t \le 2$

$\frac{Ldi}{dt} = 10 \quad \text{for } 2 \le t \le 4$

$\frac{Ldi}{dt} = 20 - 25t \quad \text{for } 4 \le t \le 8$

12. BD

Sol. $eV_0 = hr - \phi$

SECTION – C

13. 00036.00

Sol. $\frac{dNx}{dt} = 64 \times \frac{dNy}{dt} \text{at} \quad t = 0$

$\lambda_x.N_x = \lambda_y.N_y \times 64$

after time t $\lambda_x.N_x.e^{-\lambda_x.t} = \lambda_y.N_y.e^{-\lambda_y.t}$ a

On solving we get t = $\frac{\ln 64}{\ln 2\left(\frac{1}{5} - \frac{1}{30}\right)} = 36$ days

14. 00020.00

Sol. $\frac{1}{2}Kx_{max}^2 = \frac{1}{2}\mu v^2 \qquad (\mu \text{ is reduced mass})$

$x_{max} = v.\sqrt{\frac{\mu}{k}}$

$\mu = \frac{200}{3}\text{gm}$

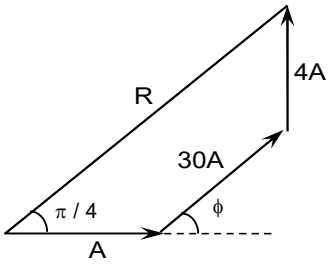
15. 00001.80

Sol. $\frac{GM_1}{a^2} = \frac{G(M_1 + M_2)}{(2a)^2} = \frac{G(M_1 + M_2 + M_3)}{(3a)^2} = g_0$

$g = \frac{GM_3}{(3a)^2}$

16. 00015.00

Sol. $\tan(\pi/4) = \frac{R_y}{R_x}$



$\Rightarrow \phi = \frac{\pi}{12}$ or 15°

17. 00001.80

Sol. $C_{eq} = \dfrac{\dfrac{30}{11}\times 5}{\dfrac{30}{11}+5} = \dfrac{150}{85}$

$Q = \dfrac{150}{85}.E;\ V_A - V_C = \dfrac{Q}{(30/11)} = \dfrac{150E\times 11}{85\times 30} = 11$ volt.

E = 17 volt.

$V_I - V_F = \dfrac{q}{7}$

18. 00140.00

Sol. $\dfrac{G\left(\dfrac{4}{3}\pi r^3\rho\right)}{r^2} > \omega^2 r \Rightarrow \rho > 1.4\times 10^{11}\ kg/m^3$

Chemistry

PART – II

SECTION – A

19. D

Sol. $r = k_3[x][A]$

$= k_3 \times \frac{k_1}{k_2} \times [A][B][A]$

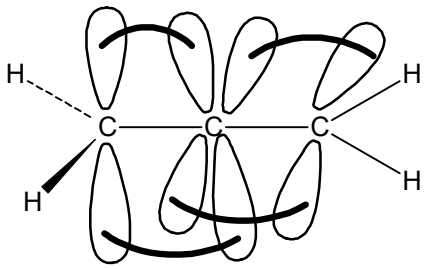
$= \frac{k_1 k_3}{k_2}[A]^2[B]$

$\therefore K = \frac{[x]}{[A][B]} = \frac{k_1}{k_2}$

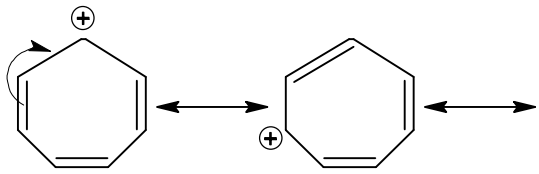
$[x] = \frac{k_1}{k_2} \times [A][B]$

20. C

Sol.



O=C=C=C=O Linear molecule



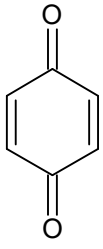
5 more resonating structure

21. C

Sol. Theory Based.

22. C

Sol.

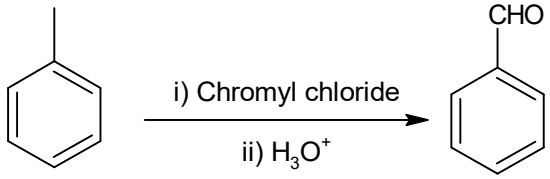


(x)

23. A

Sol. X = chromyl chloride

Y = toluene



(Etard reaction)

24. C

Sol. The blue colour of the solution of alkali metal in liquid ammonia is due to ammoniated electrons.

25. BC

Sol. Theory based.

26. ACD

Sol. $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$

$\frac{1}{\lambda} = R \times (4)^2 \times \left(\frac{1}{4^2} - \frac{1}{1^2}\right)$

$\frac{1}{\lambda} = R \times 16 \times \frac{-15}{16}$

$-\lambda = \frac{1}{15R}$

$E_n = -13.6 \times \frac{4^2}{4^2}$

$= -13.6 \text{ eV}$

$= -2.176 \times 10^{-18} \text{ Joule}$

$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$

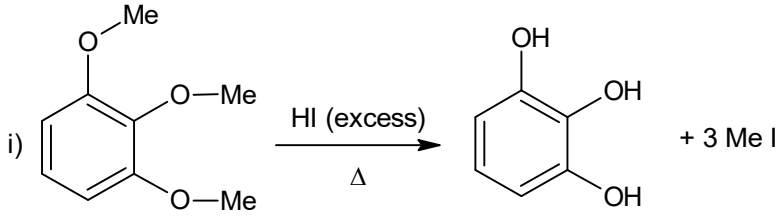
$\frac{1}{\lambda} = R \times 4^2 \times \left(\frac{1}{4^2} - \frac{1}{5^2}\right)$

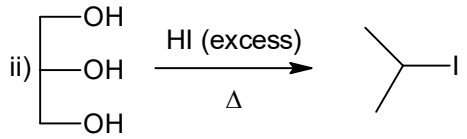
$\frac{1}{\lambda} = R \times 16 \times \frac{9}{16 \times 25}$

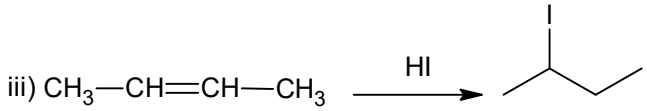
$\lambda = \frac{25}{9R}$

27. AC

Sol.





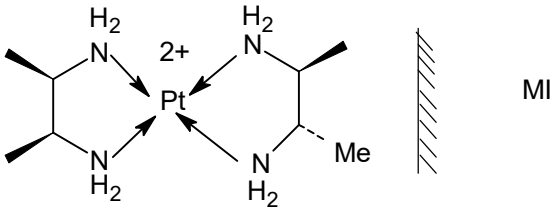


28. ABC

Sol. Theory based.

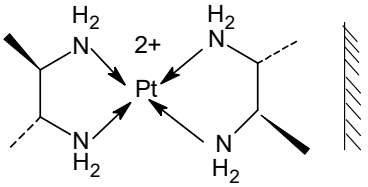
29. BC

Sol.



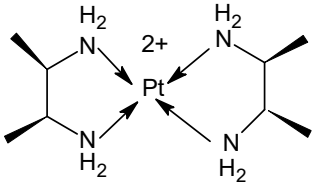
MI

Enantiomer Pair

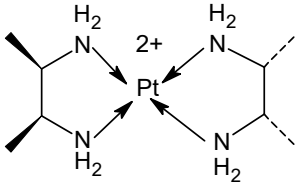


MI

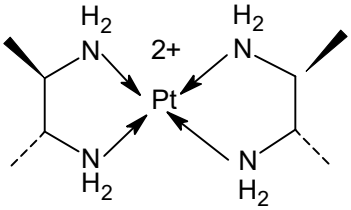
Enantiomer Pair



Meso



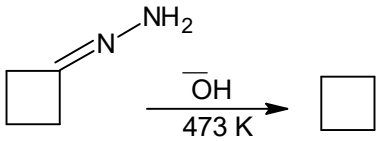
Meso



Meso

30. ABC

Sol.



SECTION – C

31. 00000.1182

Sol. LHS : $Ag \longrightarrow Ag^+ + e^-$

$\left[Ag^+\right]\left[Br^-\right] = 3\times10^{-13}$

$\left[Ag^+\right] = \dfrac{3\times10^{-13}}{5\times10^{-3}} = 6\times10^{-11}M$

RHS $Ag^+ + e^- \longrightarrow Ag$

$\left[Ag^+\right]\left[Cl^-\right] = 6\times10^{-10}$

$\left[Ag^+\right] = \dfrac{6\times10^{-10}}{1\times10^{-1}} = 6\times10^{-9}M$

$E_{cell} = \dfrac{-0.0591}{1}\log_{10}\dfrac{6\times10^{-11}}{6\times10^{-9}}$

$= -0.0591\times(-2)$

$= 0.1182$

$= 1.182\times10^{-1}$

32. 00011.00

Sol. X = 6

Y = 5

33. 00002.00

Sol. 2 (v and vii)

34. 00004.00

Sol. 4 (iii, iv, v and vi)

35. 00009.00

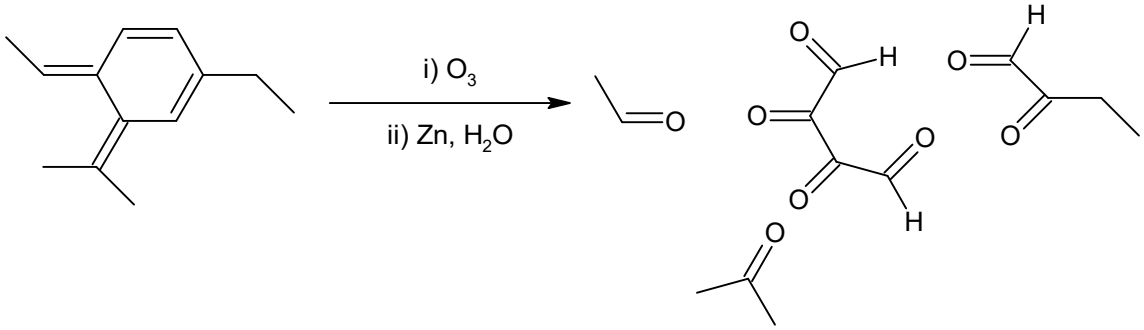
Sol. P.O.S in $SiCl_4$ is 6.

POS in ethene is 3.

$\dfrac{x^2-y^2}{(x-y)} = x+y = 6+3 = 9$

36. 00001.00

Sol.



Mathematics

PART – III

SECTION – A

37. B

Sol. $\because 2\cos\theta_1 = \frac{8}{5} \Rightarrow \cos\theta_1 = \frac{4}{5} \Rightarrow \tan\frac{\theta_1}{2} = \sqrt{\frac{1-\frac{4}{5}}{1+\frac{4}{5}}} = \frac{1}{3}$ and $e = \frac{1}{2}$

$\therefore \tan\frac{\theta_2}{2} \times \frac{1}{3} = \frac{-\frac{1}{2}}{\frac{3}{2}}$

$\frac{\theta_2}{2} = \frac{3\pi}{4} \Rightarrow \theta_2 = \frac{3\pi}{2}$

$\therefore B = \left(0,\ -\sqrt{3}\right) \Rightarrow AB = 2 + 2 - \frac{1}{2} \times \frac{8}{5} = 4 - \frac{4}{5} = \frac{16}{5}$

38. D

Sol. $\because$ Image of (2, 1) in $2x - y + 1 = 0$ is

$\frac{x-2}{2} = \frac{y-1}{-1} = \frac{-2(4-1+1)}{4+1} = \frac{-8}{5}$

$\Rightarrow x = 2 - \frac{16}{5} = \frac{-6}{5}$

$y = \frac{8}{5} + 1 = \frac{13}{5}$

$\therefore$ Slope of directrix $= \frac{\frac{13}{5} - 3}{\frac{-6}{5} - 1} = \frac{-\frac{2}{5}}{-\frac{11}{5}} = \frac{2}{11}$

$\therefore$ Equation of axis,

$y - 1 = \frac{-11}{2}(x - 2)$

$\Rightarrow 2y - 2 = -11x + 22$

$\Rightarrow 11x + 2y = 24$

39. D

Sol. $\because \left(\hat{a} \times \hat{b}\right).\left\{\left(\hat{a}.\hat{c}\right)\hat{b} - \left(\hat{a}.\hat{b}\right)\hat{c}\right\} = 0 - \left(\hat{a}.\hat{b}\right)\left[\hat{a}\ \ \hat{b}\ \ \hat{c}\right]$

$= -\frac{1}{2} \times \frac{1}{\sqrt{2}}$

$\therefore \left[\hat{a}\ \ \hat{b}\ \ \hat{c}\right]^2 = \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{1}{2}$

40. D

Sol. $\because a_1^2+b_1^2+c_1^2+a_2^2+b_2^2+c_2^2+a_3^2+b_3^2+c_3^2=7$, where $\langle a_i,\ b_i,\ c_i\rangle$ are elements of ith row.

$\therefore$ Case I: All seven entries are 1 and rest of all zero

${}^9C_7 = {}^9C_2 = 36$

Case II: Only one entry is 2 & three entries are 1 and rest of all 0's.

${}^9C_4 \times {}^4C_1 = 504$

$\therefore$ Total = 540

41. C

Sol. $\because \frac{1}{4}\sin 3\theta = \frac{3}{4}\sin\left(\frac{\theta}{9}\right) - \sin^3\theta - 3\sin^3\frac{\theta}{3} - 9\sin^3\left(\frac{\theta}{9}\right)$

$\Rightarrow \sin 3\theta = \frac{1}{\sqrt{2}} = \sin\frac{\pi}{4}$ or $\sin\frac{3\pi}{4}$

$\Rightarrow \theta = \frac{\pi}{12}$ or $\frac{\pi}{4}$

$\therefore \tan\frac{\pi}{12} = 2-\sqrt{3}$ or $\tan\frac{\pi}{4} = 1$

$\therefore \tan\frac{\pi}{12} + \tan\frac{\pi}{4} = 3-\sqrt{3}$

42. A

Sol. $a_{11} = f\left(x^{2-1-2}\right) = f\left(\frac{1}{x}\right)$, $a_{21} = f\left(x^{4-1-2}\right) = f(x)$, $a_{12} = f\left(x^{2-2-2}\right) = f\left(\frac{1}{x^2}\right)$ and $a_{22} = f\left(x^{4-2-2}\right) = f(1)$.

$$\therefore A = \begin{pmatrix} f\left(\frac{1}{x}\right) & f\left(\frac{1}{x^2}\right) \\ f(x) & f(1) \end{pmatrix} \text{ and } B = \begin{pmatrix} f(1) & f\left(\frac{1}{x}\right) \\ f\left(x^2\right) & f(x) \end{pmatrix}$$

$$\therefore A + B^T = \begin{pmatrix} f\left(\frac{1}{x}\right)+f(1) & f\left(\frac{1}{x^2}\right)+f\left(x^2\right) \\ f(x)+f\left(\frac{1}{x}\right) & f(1)+f(x) \end{pmatrix}$$

From given matrix,

$f\left(\frac{1}{x}\right)+f(1) = -x^2 f\left((x)+f(1)\right)$ ..... (i)

$f\left(\frac{1}{x^2}\right)+f\left(x^2\right) = \left(1-\frac{1}{x^4}\right)f\left(\frac{1}{x^2}\right)$ ..... (ii)

$f(x)+f\left(\frac{1}{x}\right) = f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right)$ ..... (iii)

$f(1)+f(x) = -\frac{1}{x^2}f\left(\frac{1+f(1)}{x}\right)$ ..... (iv)

From (ii), $x=1$ $\quad 2f(1)=0 \Rightarrow f(1)=0$

$\therefore$ Eqn. (i), $-x^2 f\left(x\right) = f\left(\frac{1}{x}\right)$ ..... (*)

Now, $\int_x^{1/x} f\left(t\right)dt = I$

Replace $t \to \frac{1}{t}$

$\Rightarrow I = \int_{1/x}^{x} f\left(\frac{1}{t}\right)\left(-\frac{1}{t^2}\right)dt = \int_x^{1/x} f\left(\frac{1}{t}\right)\frac{1}{t^2}dt$

$\Rightarrow 2I = \int_x^{1/x}\left(f\left(t\right)+\frac{1}{t^2}f\left(t\right)\right)dt = 0$ from (*)

i.e., $I = 0 = f\left(1\right)$

Option (a) is correct.

43. BCD

Sol. $\because \left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] = 2$ and $3\left(\bar{a}+\bar{b}\right).\left\{\left(\bar{b}+\bar{c}\right)\times\left(\bar{c}+\bar{a}\right)\right\}$

$= 3\left(\bar{a}+\bar{b}\right).\left\{\bar{b}\times\bar{c}+\bar{b}\times\bar{a}+0+\bar{c}\times\bar{a}\right\}$

$= 3\left\{2\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]\right\} = 6\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] = 12$

$\left[\bar{a}\times\bar{b}\ \ \bar{b}\times\bar{c}\ \ \bar{c}\times\bar{a}\right] = \left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]^2 = 4$

$\Rightarrow \left(\bar{a}-2\bar{b}\right).\left\{\left(\bar{b}-2\bar{c}\right)\times\left(\bar{c}-2\bar{a}\right)\right\} = \left(\bar{a}-2b\right)\left\{\bar{b}\times\bar{c}-2\bar{b}\times\bar{a}+4\bar{c}\times\bar{a}\right\}$

$= \left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]-8\left[\bar{b}\ \ \bar{c}\ \ \bar{a}\right] = -7\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]$

$\left(3\bar{b}-\bar{c}\right)\times\left(3\bar{c}+\bar{a}\right) = 9\bar{b}\times\bar{c}-3\bar{b}\times\bar{a}+\bar{c}\times\bar{a}$

$\left(3\bar{a}-\bar{b}\right).\left\{3\bar{c}+\bar{a}\right\} = 27\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]-\left[\bar{b}\ \ \bar{c}\ \ \bar{a}\right] = 26\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]$

44. BC

Sol. $\because \sqrt{6}x\sec\theta-\sqrt{3}y\,\mathrm{cosec}\,\theta = 3$ is equation of normal,

then, slope $= \frac{-\sqrt{6}\sec\theta}{-\sqrt{3}\,\mathrm{cosec}\,\theta} = \sqrt{2}\tan\theta$

Given, $\sqrt{2}\tan\theta = +1$

$\Rightarrow \tan\theta = \frac{1}{\sqrt{2}}$

$\therefore$ normal is $x-y=1$

Let $P\left(h,\ \frac{2}{h}\right)$

$\Rightarrow h-\frac{2}{h} = 1$

$\Rightarrow h^2-h-2 = 0$

$\Rightarrow h = 2$ or -1

45. ABD

Sol. $\because y = mx - 2m - m^3$

$\Rightarrow 6 = 9m - 2m - m^3$

$\Rightarrow m^3 - 7m + 6 = 0 \Rightarrow m = 1, +2, -3$

$\therefore$ Centre is $\left(\frac{11}{2}, \frac{3}{2}\right)$

46. ABC

Sol. $\because f(x) = 2 \times \max\left\{|x-1| + |x-3|,\ |2x-1|\right\}$

$\therefore x = 1,\ \frac{3}{2}$ point of non-differentiability and minimum value is 2

47. ABD

Sol. $\because \int \cos(2017x + x)\sin^{2016} x\ dx = \int \cos 2017x \sin^{2016} x \cos x\ dx - \int \sin 2017x \sin^{2017} x\ dx$ ..... (i)

$\left\{\because \int \cos \underset{(I)}{2017x} \underbrace{\sin^{2016} x \cos x}_{(II)}\ dx = \frac{\cos 2017x \sin^{2017} x}{2017} + \int \frac{2017 \sin 2017x}{2017} \sin^{2017} x\ dx\right\}$

$\therefore \int \cos 2018x \sin^{2017} x\ dx = \frac{1}{2017} \cos 2017x \sin^{2017} x$

48. BD

Sol. $\because f(x) = \frac{xf'(x)}{2\sqrt{f(x)}} + \sqrt{f(x)}$

$\Rightarrow \sqrt{f(x)} = \frac{1}{1-cx}$

$\Rightarrow f(x) = \frac{1}{(1-cx)^2}$

$\because f(1) = \frac{1}{2} \quad \therefore c = 1 - \sqrt{2}$

$\Rightarrow f(x) = \frac{1}{\left[1 + \left(\sqrt{2} - 1\right)x\right]^2}$

SECTION – C

49. 00003.00

Sol. $\because a^2 - (b+c)^2 + xbc = 0$

$\therefore x = \frac{(a-b-c)(a+b+c)}{bc}$

$= \frac{4s(s-a)}{bc} = 4\cos^2 \frac{A}{2} \in (0,\ 4)$

$\because 0 < A < \pi$

$\Rightarrow 0 < \frac{A}{2} < \frac{\pi}{2}$

50. 00000.80

Sol. $\frac{x-2}{\frac{1}{-\sqrt{2}}} = \frac{y-1}{\frac{1}{\sqrt{2}}} = 4\sqrt{2}$ or $-4\sqrt{2}$

$\therefore A(-2, 5)$ & $B(6, -3)$

$\therefore$ area of $\Delta CAB = 20$

$\Rightarrow$ sum $= 2\sqrt{20} = 4\sqrt{5}$

51. 00004.00

Sol. $\sqrt{2\left(a^2+b^2\right)} = 4 \Rightarrow a^2+b^2 = 8$

$\therefore \frac{a^2+b^2}{2} \geq ab$

$\Rightarrow ab \leq 4$

52. 00000.33

Sol. $\because x^2+5x-6=0$

$\Rightarrow (x-1)(x+6)=0 \Rightarrow x=1$

$\therefore x+5y=6$ ..... (i) tangents

& $y=\frac{1}{2}(x+1)$ ..... (ii)

$\therefore \tan\theta = \left|\frac{-\frac{1}{5}+\frac{1}{2}}{1-\frac{1}{10}}\right| = \frac{3}{9} = \frac{1}{3}$

53. 00002.00

Sol. $\frac{a+b}{2} = \frac{5}{4}\sqrt{ab}$

$\Rightarrow \frac{a}{b} = \frac{1}{4}$

$\frac{H_8-a}{a-H_8} = 8\times\frac{a}{b} = 8\times\frac{1}{4} = 2$

54. 00002.00

Sol. $\because f(x) = \sum_{i=1}^{\infty}\left[\frac{1}{(i-1)x+1} - \frac{1}{ix+1}\right]$

$\Rightarrow f(x) = \lim_{n\to\infty}\left[1-\frac{1}{nx+1}\right]$

$\Rightarrow f(x) = 1$

$\therefore f\left(\frac{1}{x}\right) = 1$

$\Rightarrow f(x)+f\left(\frac{1}{x}\right) = 2$